

WhatABook Design

Building a Python application prototype, connecting readers to WhatABook library databases in the comfort of their homes.

PURPOSE:

In our daily lives, whether within our digital environment, or going to places, consumers are exposed to an overwhelming amount of information, making it so difficult to filter content on what they need or desire to be informed on, making it so hard to make decisions.

According to a 2009 study by researchers at the University of California, San Diego, "Americans consumed an average of about 100,000 words worth of information per day, spanning television, computers, radio, and other media." (Bohn & Short, 2009).

This challenge becomes evident when searching for books online. With the presence of paid advertisements, promotional content, or algorithm driven recommendations, finding a book, or using set of keywords to locate good and reliable source of information that aligns with one's interests can require lots time and mental power...

That's why WhatABook wants to address this issue by simplifying the discovery process.

By Building a super structured Python application connected to a MongoDB database, this system will allow users to efficiently browse books by genre, author, title, or bookId, and manage personalized wishlists. By organizing information in a clear and accessible way, WhatABook will allow its users to bring this system home, so that they can browse and easily locate books in the confort of their homes.

Finally, this will be a Python based system that connects to a MongoDB database. Python will act as the connector that interacts directly with the user, handling inputs and displaying results in a clear format that connects to a MongoDB database to organize and manage book and customer information.

Reference:

Bohn, R., & Short, J. (2009). *How much information? 2009 report on American consumers.* University of California, San Diego.
https://group47.com/HMI_2009_ConsumerReport_Dec9_2009.pdf

PERSONAS:

Persona 1 - Description

Lucas Andrade is a 38-year-old elder millennial who recently transitioned into a higher-paying technology role after years of balancing creative work and financial instability. As a lifelong musician, Lucas has developed a growing interest in the intersection of music, technology, and human movement. Now that he has greater financial stability and more structured free time, he is investing in books that explore science, body mechanics, sensory systems, music technology, and innovation.

Characteristics

Age: 38

Generation: Elder Millennial

Occupation: Finance Technology professional with a creative background

Hobbies: Music production, experimenting with motion sensors and sound, reading about neuroscience and tech

Income Level: Recently increased disposable income

Primary Goal: Find reliable books about science, technology, and body movement integration

Most Used Feature: Search by genre and author + wants to explore, needs to search by specific keywords

Least Used Feature: Simple "browse all books" listing

Needs the Application Can Meet: Filter books efficiently + Avoid online advertisement noise + Discover niche topics related to music and technology

Computer Skill Level: Intermediate to Advanced

Motivation: Personal growth and creative exploration

Frustration Points:

Algorithm driven ads -- can't stand them!

Overwhelming search results

Lack of focused filtering

Stories:

- As a tech professional and musician, I need to browse books by genre, so that I can easily locate books related to science, technology, and music innovation.
- As a worker with limited free time after work, I need to search for books by specific keywords, so that I can efficiently find specific books without scrolling through many results.
- As a reader who values trusted sources, I need to view books by author, so that I can explore more works from writers I trust.
- As a returning customer, I need to add books to my wishlist, so that I can save titles for later. As a returning customer, I need to add books to my wishlist, so that I can save titles for later.

- As an organized user, I need to view my wishlist, so that I can manage my selected books.

Persona 2 - Description

Persona: Emily Carter

Emoji/Icon: 📖

Profile

Emily Carter is a 32-year-old middle-school English teacher who uses WhatABook to quickly find, save, and organize books for her students and her own reading lists.

Characteristics

Name: Emily Carter

Age: 32

Occupation: English teacher

Distinguishing characteristics impacting design:

Frequently multitasks; needs simple, distraction-free UI

Uses mobile devices during class prep

Prefers fast search and filtering

Most-used features:

Book search

Genre filtering

Favorites list

Book summaries

Least-used features:

Admin book deletion

System configuration settings

Needs the app can meet:

Quickly find age-appropriate books

Maintain curated reading lists

Save books to review later

Computer skill level: Intermediate

Additional elements: **Motivation:** Inspire students to read

Pain points: Limited time, overwhelmed by too many book sources

Preferred device: iPad and school laptop

Environment: Busy classroom, short planning periods

User Stories

WhatABook includes user accounts, book search, favorites, and admin tools.

Users can browse without logging in, but saving favorites requires an account.

Admins can delete outdated books.

The book catalog includes metadata (title, author, genre, summary).

Persona 3 – Description

Persona: Robert Martinez

Emoji/Icon: 📖

Robert Martinez is a 55-year-old small business owner who enjoys reading historical fiction and biographies in his free time. He is not highly technical and prefers simple systems that are easy to navigate

Age: 55

Occupation: Small business owner

Distinguishing characteristics impacting design:

Prefers large, clear text

Not techsavvy

Does not like too many options at once

Most used features:

Browse by genre

Search by title

View wishlist

Least used features:

Advanced keyword filtering

Author exploration tools

Needs the app can meet:

Simple navigation

Clear book listings

Easy wishlist management

Computer skill level: Basic to Intermediate

Additional elements:

Motivation: Relaxation after work Pain points: Confusing layouts and too many search results

Preferred device: Desktop computer at home

Environment: Quiet home offic

User Stories:

- As a reader, I need to browse books by genre, so that I can not waste time.
- As a user with basic computer skills, I need to search by title, so that it's easy for me.
- As a customer, I need to view book details, so that I can decide if I want to read it.
- As a returning user, I need to add books to my wishlist, so that I can save them for later.
- As an organized reader, I need to view my wishlist, so that I can manage my selected books.

Work Estimation

Table: Stories , Tasks , Hours , Story Points

ID	Story / Task	Work Hours	Story Points
S1	As a teacher, I need to search for books by genre so that I can quickly find titles for my students.	—	5
T1.1	Design genre search UI	3 hrs	2
T1.2	Implement backend genre filter	4 hrs	3
T1.3	Connect UI to API	3 hrs	2
T1.4	Add validation and error handling	2 hrs	1
T1.5	Write unit tests	3 hrs	2
S2	As a reader, I need to save favorite books so that I can easily return to them later.	—	8
T2.1	Create favorites database schema	3 hrs	2

T2.2	Build “Add to Favorites” button	4 hrs	3
T2.3	Implement favorites API endpoints	4 hrs	3
T2.4	Display favorites list UI	3 hrs	2
T2.5	Write integration tests	3 hrs	2
S3	As a new user, I need to create an account so that I can save my book preferences.	—	5
T3.1	Build registration form	3 hrs	2
T3.2	Implement password hashing	2 hrs	1
T3.3	Create user registration API	4 hrs	3
T3.4	Add form validation	2 hrs	1
T3.5	Write unit tests	3 hrs	2
S4	As a returning user, I need to update my profile so that my account information stays current.	—	3
T4.1	Create profile update UI	3 hrs	2
T4.2	Build update API endpoint	3 hrs	2
T4.3	Add input validation	2 hrs	1
T4.4	Implement success/error messages	1 hr	1
T4.5	Write tests	2 hrs	1
S5	As an admin, I need to delete outdated book records so that the catalog remains accurate.	—	8
T5.1	Build admin authentication check	4 hrs	3
T5.2	Create delete-book API endpoint	4 hrs	3

T5.3	Add confirmation modal UI	2 hrs	1
T5.4	Implement audit logging	3 hrs	2
T5.5	Write admin-level tests	3 hrs	2

SITE MAP:

The sitemap was inspired by structured film and music libraries, where users can browse, search, and filter content in intuitive and layered ways. I wanted WhatABook to feel less like a basic catalog and more like a curated digital library experience. The navigation separates browsing, searching, and filtering into clear categories so users can move intentionally instead of feeling overwhelmed. Wishlist features are grouped independently to support personal organization and allow users to build and manage their own reading collections in a focused and structured way.

Sitemap – WhatABook Console Application

WhatABook Console Application

1. Main Menu

1.1 Browse & Filter Books

1.1.1 View All Books

1.1.2 Browse by Genre

1.1.3 Browse by Author

1.1.4 Browse by Style

1.1.5 Browse by Theme

1.1.6 Filter by Difficulty Level

1.1.7 Filter by Publication Era

1.1.8 Staff Picks

1.2 Search Books

1.2.1 Search by Title

- 1.2.2 Search by Book ID
- 1.2.3 Search by Keyword

1.3 Wishlist

- 1.3.1 View Wishlist
- 1.3.2 Add Book to Wishlist
- 1.3.3 Remove Book from Wishlist

1.4 Events

- 1.4.1 View Upcoming Events
- 1.4.2 Event Details

1.5 Blog

- 1.5.1 View Recent Posts
- 1.5.2 View Post Details

1.6 Contact

- 1.6.1 Submit Inquiry
- 1.6.2 View Store Information

1.7 Exit Application

Prototype

Marvel Prototype Link: <https://marvelapp.com/prototype/83gi97e>

UI Principles / Best Practices Applied (2+)

Simplicity + reduced cognitive load: numbered menus, one decision per screen, minimal text, predictable prompts—aligned with the requirement that the interface be “as simplistic as possible.”

Consistency: every results screen uses the same book row format and always offers a clear return path (e.g., “Press Enter to return to menu”), improving learnability for novice users.

Error prevention + recovery: constrained input (numeric menu choices; known customerIds) and friendly error messages for invalid customerId (explicitly required).

